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Brazilian Coffee Consumers: Behavioral Insights From a Leading Global Coffee Market

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ABSTRACT

Brazilian coffee consumers were profiled through an online survey (1200 participants), analyzing preferences, habits, and attitudes. Behaviors of sweetened (SC) and unsweetened (UC) coffee drinkers were compared. Results reveal high coffee consumption rooted in family habits, tradition, socialization, and daily routines. A greater appreciation for differentiated coffees and preparation methods was observed, particularly among self-identified enthusiasts. Additional trends included increased demand for medium-roast, certified, gourmet, and specialty coffees, a rise in espresso, and growing purchases through online platforms and coffee shops. However, some consumers remain unaware of the health benefits of coffee ingestion, highlighting need for broader dissemination of information. Both groups value flavor and health aspects, but UC stood out for reduced food neophobia. While sugar use decreased among SC, UC consumers seemed more focused on quality and innovation, exploring preparation methods and emphasizing attributes such as roasting degree and origin. Findings highlight the coexistence of tradition and innovation in Brazilian coffee culture, providing insights for product development, marketing, and public health communication.

1 | Introduction

Coffee is the second most-consumed beverage in the world. In the 2024/2025 harvest year, global coffee production reached almost 175 million 60-kg bags. Brazil, the world's largest producer, accounted for 37% of this total and was also the second-largest consumer of coffee (USDA 2025).

In Brazil, most information on the profiles and opinions of coffee consumers comes from surveys commissioned by private consultancies and made available by the Brazilian Coffee Industry Association (ABIC); this institution is part of the Coffee Policy Deliberative Council/Mapa, which represents the coffee roasting and milling industries in the country. The surveys stand out for their comprehensive nature, featuring many respondents and representing consumers from different Brazilian regions. Over the years, these surveys have shown that the Brazilian coffee market is dynamic, with room for innovation and continuous growth (Axxus 2025), and have described significant changes in consumer habits and preferences (Axxus 2025; Euromonitor 2015, 2016, 2017; Ivani Rossi 2010; Maia et al. 2014; Nielsen 2013; SPCH-ABIC 2021; Universidade do Café 2025).

Between 2003 and 2010, Ivani Rossi (2010) conducted annual surveys on coffee consumption trends. These surveys highlighted several changes, including the emergence of specialty coffee consumption, increased demand for decaffeinated products, a rise in consumption among young people and the lower-income class C, and a growing trend of out-of-home coffee consumption. However, since 2014, due to Brazil's economic crisis, the frequency and scope of these studies have declined,

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Summary

The findings provide relevant insights into the distinct behavioral patterns of sweetened and unsweetened coffee consumers and help to understand shifting consumer preferences in the largest coffee markets. Considering the high level of coffee consumption in Brazil, these results can support the coffee sector in developing targeted educational campaigns to promote products with reduced or no added sugar. In addition, the findings highlight opportunities for improved labeling strategies, including clearer communication about the benefits of regular coffee consumption. Differences in consumption contexts and preparation methods also suggest that product formats and convenience-oriented solutions can be tailored to specific consumer profiles. Overall, these findings can support stakeholders across the coffee value chain in designing products and strategies that better align with evolving consumer preferences and sensory expectations.

resulting in limited information on the evolution of consumer profiles over the past decade. Moreover, the contracting of different research agencies introduced variations in both the topics addressed and the methodologies used, making it difficult to compare data over time.

Despite these challenges, some key trends can be observed. These include an increasing demand for higher-quality and specialty coffees, reflecting a more discerning and informed consumer, as well as pandemic-related shifts, such as the intensification of at-home coffee preparation and the expansion of digital sales platforms (Axxus 2025; SPCH-ABIC 2021). These developments merit further monitoring to assess how the market continues to adapt and consolidate.

In parallel with the research conducted by consultancies, several academic studies have examined the behavior of Brazilian coffee consumers regarding specific topics. These studies, however, were conducted with smaller sample sizes than the present study and pursued diverse objectives. Some explored the motivations behind coffee consumption or rejection (Arruda et al. 2009; Costa et al. 2024; Sousa and Costa 2015; Sousa et al. 2016), while others focused on sensory acceptance and preferences between unsweetened (UC) and sweetened (SC) coffee beverages, as well as the influence of consumer characteristics on consumption patterns (Fonteles et al. 2010; Moraes and Bolini 2010; Portela et al. 2024; Souza et al. 2013; Ued et al. 2024).

Additional studies investigated the impact of roasting degree, processing methods, and quality on consumer acceptance (Bressani et al. 2021; Monteiro et al. 2010; Viencz et al. 2024). Research also addressed purchasing behavior and discussed survey data on consumption trends (Siqueira et al. 2021; Soares, Ferreira, et al. 2023; Sousa and Costa 2015). Changes in consumption patterns during the COVID-19 pandemic were explored, along with the consumption potential of various preparation methods among young Brazilians (Violin et al. 2020).

One of the points identified in these studies is the widespread practice of adding sugar to coffee brews, a common practice in Brazil (Wan-Mohtar et al. 2025). Despite its relevance, this topic has been underexplored in broader research, and methodological differences among studies have made it difficult to track changes in consumer behavior over time.

In recent years, the rise in obesity rates and excessive sugar consumption—above the levels recommended by the World Health Organization—has led to investigations into which foods in the Brazilian diet contribute to these issues (Brasil 2024; Ued et al. 2024). Data from the Household Budget Survey (POF 2017–2018) indicate that sugar is more often introduced into meals through coffee preparation (Siqueira et al. 2021; Souza et al. 2013; Ued et al. 2024).

Strategies to reduce added sugar in coffee brews without compromising sensory acceptance have already been proposed (Portela et al. 2024). This context highlights the need for detailed information on the behavior, habits, and preferences of consumers of SC and UC coffees, which may contribute to addressing the broader public health challenge.

In this context, the present study aimed to provide an updated profile of Brazilian coffee consumers regarding their preferences, habits, and awareness, particularly regarding the health benefits of coffee, and investigating how these aspects influence the consumption of SC and UC coffee beverages.

2 | Material and Methods

2.1 | Sampling and Data Collection

This study was approved by the Research Ethics Committee of the Universidade Estadual de Londrina (CAAE 57783722.7.0000.5231).

With the aim of obtaining a representative sample of the Brazilian population (163 million inhabitants aged 15 or older), a questionnaire was distributed in proportion to the number of inhabitants in each region of the country, using information from the 2022 Census conducted by the Brazilian Institute of Geography and Statistics (IBGE 2022). Considering a 95% confidence level and a 5% margin of error, the sample calculation indicated that at least 384 responses were needed (Equation 1). To increase the robustness of the results, 1200 responses were collected (Table 1), maintaining the established proportion for each region, which resulted in a 99% confidence level and an error of 3.72%.

$$n = \frac{N.\hat{p}.\hat{q}.\left(Z_{\frac{\alpha}{2}}\right)^2}{\hat{p}.\hat{q}.\left(Z_{\frac{\alpha}{2}}\right)^2 + (N-1).E^2} \quad (1)$$

where n is the number of individuals by sample; N is the number of individuals in the population (163,000,000); $Z_{\alpha/2}$ is the critical value corresponding to the desired degree of confidence (2.576); $\hat{p}$ is the population proportion of individuals in the category under study (0.50); $\hat{q}$ is the population proportion of individuals

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not belonging to the category under study (0.50); and E is the margin of error (3.72%).

Data were collected via a questionnaire hosted on the Google Forms platform between November 2023 and February 2024. The questionnaire link was distributed through various channels, including email, institutional social media, messaging apps, and in-person at two industry events in 2023 (International Coffee Week—SIC and the 2nd Brazilian Coffee Tasters Meeting—IFES; IFES 2023; SIC 2023), to reach consumers with greater knowledge of the subject. All study participants were at least 18 years old, as stated to the Research Ethics Committee.

2.2 | Questionnaire for Data Collection

The questionnaire was initially validated with 10 participants, which allowed for the identification and correction of ambiguous or unclear questions, thereby enhancing data quality. After making the adjustments, the estimated time to complete the questionnaire was 10–20 min.

TABLE 1 | Population sampling by geographic regions of Brazil.

Region	No. of inhabitants[a]	Sampling	
		Collected answers	% Total
North	12,983,834	96	8
Northeast	43,108,045	317	26
Midwest	12,887,538	95	8
Southeast	69,559,389	512	43
South	24,412,689	180	15
Total	162,951,495	1200	100

[a]Aged 15 years and older; 2022 Census (IBGE 2022).

The questionnaire was structured into seven sections (Figure 1). Participants first reviewed the Informed Consent Form (ICF) and, upon agreeing to participate and confirming they were coffee consumers, proceeded to the survey.

Sections two and three gathered sociodemographic information and general coffee consumption habits (e.g., frequency, time, and place of consumption). Participants were also asked to classify themselves into one of three consumer profiles, based on their self-perception: general public (i.e., individuals who drink regular/ordinary coffee, are unfamiliar with specialty coffees, and do not prioritize the subject); enthusiasts (i.e., individuals with some knowledge of coffee, who purchase differentiated products, and are engaged with the topic); or experts (i.e., individuals with in-depth knowledge, who have taken courses and may or may not work in the field, but are able to discuss the topic technically). These categories and definitions were adapted from a survey conducted by the São Paulo Coffee Hub in partnership with ABIC, which aimed to characterize consumers interested in high-quality coffee (Axxus 2025; SPCH-ABIC 2021).

After participants indicated whether they consumed SC coffee beverages—including those prepared with sucrose or sweeteners—or UC coffee beverages, the questionnaire followed a differentiated structure for each group (Figure 1). Section four was directed exclusively at SC consumers and examined the reasons for their preference, as well as the amounts of sugar and/or sweetener used. Section five focused on UC consumers, investigating the motivations behind their choice.

In the sixth section, all participants were asked to provide more detailed information about their coffee consumption habits, including their knowledge of different coffee species and varieties,

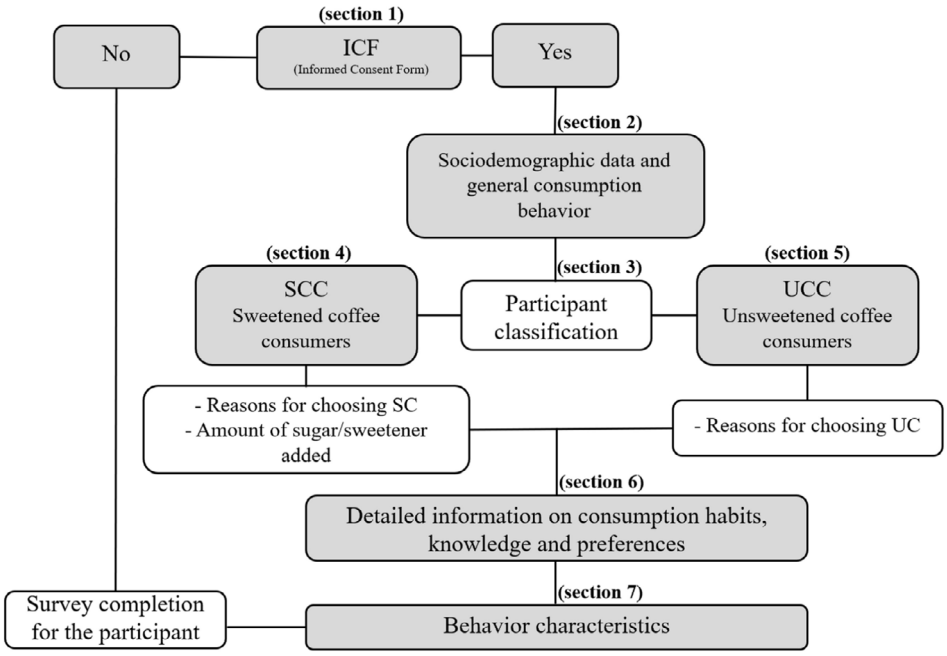


FIGURE 1 | Structure of the questionnaire developed and used in the research.

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TABLE 2 | Items included in the instrument to assess functional food acceptance.

Knowledge
1. How do you judge your personal knowledge of functional foods?
Acceptance of Functional Foods
2. Functional foods are acceptable to me if they taste good.
3. Functional foods are acceptable to me even if they taste worse than their conventional alternative foods.
Perceived Role of Food for Health
4. Food plays an important role for my personal health.
5. I feel to have control over my personal health.
6. I feel eating healthier now compared to 5 years ago.
Health Benefit Belief
7. Functional foods are likely to have a beneficial impact on my personal health.
8. I experience functional foods as being part of a natural way of living.
9. Functional foods allow me to take my personal health in my own hands.
10. Functional foods are a convenient way of meeting recommended daily intakes, which I would never meet with my conventional diet.

Source: Verbeke (2005).

the attributes they consider when purchasing coffee, their preferences regarding roasting degree and preparation methods, whether at home or elsewhere, and they were asked to provide four words they associate with quality coffee.

The seventh section addressed behavioral aspects of consumers, including food neophobia (i.e., aversion to unfamiliar foods), knowledge and acceptance of functional foods, and perceived health benefits and risks associated with coffee consumption. These topics are discussed in detail in the second section. The final section of the questionnaire aimed to assess how these factors influence coffee consumption, particularly in comparing the behaviors of SC and UC consumers.

2.3 | Evaluating the Acceptance of Functional Foods

This part of the research assessed participants' knowledge and acceptance of functional foods. Perceptions regarding the impact of food on health, as well as beliefs about the health benefits of functional foods, were evaluated using a questionnaire developed initially by Verbeke (2005) and later validated for Brazilian consumers by Corso and Benassi (2012) (Table 2).

A 5-point Likert scale was used to assess most questionnaire items, ranging from 1 (strongly disagree) to 5 (strongly agree). An exception was the item on knowledge, which was measured using a 3-point scale with the options: 'low' (1), 'medium' (2), and 'high' (3).

Two items were central to defining acceptance: 'Functional foods are acceptable to me if they taste good' (item 2) and 'Functional foods are acceptable to me, even if they taste worse than their conventional alternative foods' (item 3). Both items consider taste as a key factor in food acceptance. Responses showing disagreement (score <3) with item 2 and simultaneous agreement (score ≥ 3) with item 3 were considered contradictory and excluded from the dataset, as recommended by Verbeke (2005). As a result, 147 out of the 1200 responses were excluded from this part of the analysis.

The items related to "Perception of the role of food in health" (items 4–6) were summed to generate a continuous variable, ranging from 1 to 15, which was used in the bivariate analysis (Cronbach's alpha = 0.68). Similarly, the 'Belief in health benefits' statements (items 7–10) were summed, yielding a total score from 1 to 20 (Cronbach's alpha = 0.84).

2.4 | Food Neophobia Assessment

Food neophobia was measured using a validated questionnaire developed by Andrade Previato and Behrens (2015), which consists of 10 statements rated on a 7-point Likert scale (1 = strongly disagree to 7 = strongly agree) (Table 3). Items 2, 3, 5, 7, and 8 are positively associated with neophobia, while items 1, 4, 6, 9, and 10 reflect neophilic tendencies and were reverse scored.

For data analysis, individual scores were calculated by summing responses to all scale items, resulting in total values ranging from 10 to 70 points. Following the classification proposed by Olabi et al. (2009), participants were categorized based on the mean score (31.3) and standard deviation (9.1): scores ≤ 22.3 (mean—1 SD) indicated food neophilia; scores between 22.4 and 40.3 were considered neutral; and scores ≥ 40.4 (mean + 1 SD) indicated food neophobia.

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TABLE 3 | Items included in the instrument used to assess food neophobia.

Scale statements[a]
1. I am constantly sampling new and different foods (R)
2. I don't trust new foods
3. If I don't know what is in a food, I won't try it
4. I like foods from different countries (R)
5. Ethnic food looks too weird to eat
6. At dinner parties, I will try a new food (R)
7. I am afraid to eat things I have never had before
8. I am very particular about the foods I will eat.
9. I will eat almost anything (R)
10. I like to try new ethnic restaurants (R)

[a]7-point Likert Scale (1 = strongly disagree and 7 = strongly agree); (R): Reversed item: for statements corresponding to neophilia, scale values are reversed.
Source: Andrade Previato and Behrens (2015).

2.5 | Statistical Analysis

Data were analyzed using descriptive exploratory methods, with categorical variables expressed as percentages. As some questions allowed multiple responses, percentages in some instances exceeded 100%. Comparisons between the SC and UC groups were performed using the k-proportions test ($p \leq 0.10$) in XLStat software, version 2023 (Lumivero, New York, USA).

Responses to the prompt 'Name the first four words that come to mind when you think about a quality coffee' were categorized using a triangulation technique involving three independent researchers. Only categories mentioned by more than 5% of participants were included in the analysis.

Data on the acceptance of functional foods were analyzed using descriptive statistics. As the responses to items 2 and 3 did not follow a normal distribution (Shapiro–Wilk test), differences between SC and UC groups were evaluated using the Mann–Whitney *U* test at a 5% significance level.

3 | Results and Discussion

3.1 | General Characterization of the Participants

The survey included 1200 participants in general, and the target number of participants for each region was achieved (Table 1). Regarding the total number of participants, 59% were female and 41% male (Table 4). For comparison, women represented 52% of the Brazilian population aged 15 and over in 2022 (IBGE 2022). Most respondents were adults aged 25–55 years (68%), with notable representation from both older adults (19% over 55) and younger individuals (13% aged 18–24) (Table 4). According to the national demographic profile, the 25–54 age group also comprises the largest share (55%), followed by those aged 55 and older (27%) and 15–24-year olds (18%) (IBGE 2022). Therefore, the study sample was appropriately designed, albeit with a slightly higher proportion of women and individuals in the 25–55 age range.

Other characteristics were less representative of the general population. In terms of family income, the survey participants had a homogeneous distribution, with ranges of up to 5 minimum wages (35%), 5–10 minimum wages (32%), and over 10 minimum wages (33%) (Table 4). However, these values are considerably higher than the national average per capita income in Brazil in 2023, which was approximately 1.5 minimum wages (IBGE 2024a). Participants also exhibited a high level of education, with 19% having completed high school, 30% holding an undergraduate degree, and 51% possessing a post-graduate degree (Table 4). By comparison, in 2023, 55% of Brazilian adults aged 25 and older had completed high school (the local's mandatory basic education), while only 20% had completed higher education (undergraduate or postgraduate) (IBGE 2024b). These sample limitations—namely, the overrepresentation of individuals with higher income and education levels—stem from the data collection method, which favored the participation of consumers with greater internet access and a higher degree of interest or willingness to engage in the research. This limitation is considered in the interpretation of the study's findings.

Among the respondents, 39% identified themselves as part of the general public, 55% as enthusiasts, and 6% as experts. Notably, only 9% reported having a direct connection to the coffee sector (e.g., producers, researchers, traders, or baristas) (Table 4). Therefore, the high proportion of self-declared enthusiasts does not necessarily mean a higher level of technical knowledge about coffee but rather suggests greater purchasing power for differentiated products or a heightened interest in the subject.

In the present study, income emerged as a relevant factor in distinguishing among consumer profiles (Table 5), in contrast to findings from the São Paulo Coffee Hub and ABIC survey (SPCH-ABIC 2021). While the earlier study reported that 95% of both general public and enthusiast consumers had incomes within the 1–3 minimum wage range, the current data indicated a wider income distribution. Specifically, 48% of the general public earn 1–5 minimum wages, whereas only 28% of enthusiasts fall within this income bracket.

3.2 | Patterns and Motivations in Coffee Consumption

Most participants reported initiating coffee consumption during their school years or early working life, with 71% starting before the age of 18% and 15% between 18 and 24 years. Among those who began drinking coffee before turning 18, 68% reported starting in childhood (0–12 years old). This early onset is in line with findings in studies of young adults (e.g., start of coffee use before 14 years in some populations) (Kelly and Prichard 2016; Pressca 2017). This early start is further supported by the fact that the majority reported first consuming coffee at home (85%) or at the homes of friends and family (9%) (Table 6).

A 2014 survey (Maia et al. 2014), which examined coffee consumption habits within Brazilian households, highlighted that

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TABLE 4 | General characterization of participants classified by groups: consumers of sweetened coffee (SC) and unsweetened coffee (UC) beverages.

		General		SC		UC	
Question	**Response/answer[a]**	**1200**	**100%**	**550**	**46%**	**650**	**54%**
Gender	Male	41%		204	37%^{B}	287	44%^{A}
	Female	59%		345	63%^{A}	363	56%^{B}
	Others	0%		1	0%^{A}	0	0% ^{A}
Age	18–24 years old	13%		85	15%^{A}	68	10%^{B}
	25–39 years old	38%		186	34%^{B}	274	42%^{A}
	40–55 years old	30%		157	29%^{A}	199	31%^{A}
	Over 55 years old	19%		122	22%^{A}	109	17%^{B}
Education level	Incomplete high school	1%		8	1%^{A}	4	1%^{A}
	Complete high school	6%		55	10%^{A}	18	3%^{B}
	Incomplete undergraduate degree	12%		88	16%^{A}	61	9%^{B}
	Complete undergraduate degree	20%		116	21%^{A}	120	18%^{A}
	Incomplete Postgraduate	10%		49	9%^{A}	71	11%^{A}
	Complete Postgraduate	51%		234	43%^{B}	376	58%^{A}
Family income[b] (minimum wage per month)	1–5	35%		248	45%^{A}	176	27%^{B}
	> 5–10	32%		154	28%^{B}	227	35%^{A}
	> 10–20	21%		96	17%^{B}	159	24%^{A}
	> 20	12%		52	9%^{B}	88	14%^{A}
Which of the following best describes your role in the coffee chain?	Consumer	99%		545	99%^{A}	638	98%^{A}
	Producer	2%		4	1%^{B}	23	4%^{A}
	Researcher	4%		11	2%^{B}	36	6%^{A}
	Trader	2%		1	0%^{B}	17	3%^{A}
	Barista	1%		0	0%^{B}	16	2%^{A}
How do you define yourself as a coffee consumer?	General public	39%		309	56%^{A}	163	25%^{B}
	Enthusiast	55%		231	42%^{B}	424	65%^{A}
	Expert	6%		10	2%^{B}	63	10%^{A}

[a]Expressed in absolute numbers and percentages. Different capital letters in the same row indicate a statistical difference ($p \leq 0.10$) based on the k-proportions test between SC and UC groups.
[b]Official minimum wage in Brazil in 2023: $241 per month.

couples commonly shared the responsibility for purchasing and preparing coffee. Complementing this, other studies have shown that coffee consumption often takes place in a family setting: 86% of respondents reported drinking coffee with family members, and 61% stated that the habit began in childhood, frequently initiated by their mothers offering the beverage (Ivani Rossi 2010).

It is worth noting, however, that previous Brazilian surveys did not report specific data on the age at which individuals began drinking coffee, which limited the possibility of conducting a more detailed analysis of how drinking habits develop over the life course.

The tendency for Brazilians to begin drinking coffee in a family environment is supported by a study by Violin et al. (2020), which found that most young consumers aged 15–19 did not prepare their own coffee, highlighting the strong influence of domestic and social contexts on shaping coffee consumption habits. In contrast, a study conducted among university students in Algeria (Hamitri-Guerfi and Bachir-Bey 2024) reported an average age onset of coffee consumption of 16.5 ± 5.91 years, with regular intake typically beginning between the ages of 15 and 20. Developmental changes and social dynamics typical of adolescence likely influence this behavior. However, the earlier onset of coffee consumption in Brazil may be attributed to the country's deeply rooted coffee culture, which is intrinsically tied to national identity and everyday

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TABLE 5 | Family income of participants self-classified as general public, enthusiasts, or experts.

Family income (minimum wage/month)	General public	Enthusiasts	Experts
1–5	48%[a]	28%[b]	19%[b]
> 5–10	28%[a]	34%[a]	33%[a]
> 10–20	17%[b]	23%[a]	36%[a]
> 20	8%[b]	15%[a]	12%[ab]

Note: Official minimum wage in Brazil in 2023: $241 per month. Different letters in the same line indicate statistical difference ($p \leq 0.10$) by the k-proportions test between SC and UC.

life. This cultural familiarity likely fosters early exposure to and acceptance of the beverage. Supporting this, a recent study by Costa et al. (2024) on Brazilians' emotional and affective relationship with coffee revealed that the positive perception of the drink is strongly associated with affective memories and a sense of comfort, further reinforcing its early and enduring presence in Brazilian households.

Although the home environment plays a predominant role in initiating coffee consumption, in this study, other contexts, such as the educational setting (7%) and the workplace (17%), were also reported as places where this habit began (Table 6). These findings suggest that socialization and routine in institutional or professional settings also significantly contribute to the formation of coffee-drinking habits in Brazil.

The social dimension of coffee consumption in Brazil becomes especially evident when examining changes that occurred during the COVID-19 pandemic, a period marked by limited social interaction. In 2019, before the pandemic, 42% of respondents identified coffee consumption as an opportunity for socialization. This figure dropped sharply to just 3% in 2021, during the peak of social distancing, but rebounded to 35% in 2023, reflecting a partial return to pre-pandemic habits, and remained relatively stable in 2025 (33%) (Axxus 2025).

The significance of coffee as a social connector also appears to vary across age groups. When asked to describe what coffee represented to them, younger consumers (up to 24 years old) emphasized its role in facilitating interaction, while individuals aged 25–54 associated it more with mood enhancement. In contrast, those over 54 viewed coffee consumption primarily as a ritual (Axxus 2019). These generational differences highlight the multi-faceted meanings associated with coffee in Brazilian society, shaped by both personal experiences and broader cultural patterns.

Regarding frequency and timing, most participants (73%) reported consuming coffee two or more times per day, typically during meals, such as breakfast (82%), afternoon snack (47%), and after lunch (41%), as well as between breakfast and lunch (37%) (Table 6). This pattern aligns with findings from a 2021 survey (SPCH-ABIC 2021), in which 39% of respondents reported preparing coffee twice a day, followed by 24% who prepared it three or more times, and 13% who prepared it only once a day.

The frequent use of thermoses, cited by 67% of participants, suggests that coffee prepared earlier in the day is often consumed later in the day (Maia et al. 2014). In 2015, the average daily coffee intake in Brazil was estimated at 3.7 cups per person (Euromonitor 2015). In the last 5 years, surveys indicated that 69%–75% of consumers reported drinking more than three cups per day (Axxus 2025; Universidade do Café 2025). According to the classification proposed by Simon et al. (2022), in which 0.5–3 cups per day is considered light to moderate consumption and more than three cups is considered high, these results suggest that Brazilian consumers typically fall within the moderate to high consumption range.

The predominance of coffee consumption at breakfast (Table 6) corroborates what has historically been reported in the literature. Surveys with Brazilian consumers reported a frequency pattern related to the time of coffee consumption: the predominance during the morning—at breakfast—often exceeding 90% of responses—followed by afternoon snacks, after lunch, and mid-morning periods (Axxus 2025; Euromonitor 2015; Universidade do Café 2025). A similar profile was observed in the present study (Table 6), with some variations in the order of frequency. Interestingly, among those who reported preparing coffee only once a day, the majority indicated doing so in the morning (SPCH-ABIC 2021), reinforcing the centrality of breakfast in national coffee habits.

Evening consumption, on the other hand, presents a contrasting tendency. While previous surveys reported that 22% to 38% of respondents consumed coffee after dinner or at night (Axxus 2025; Euromonitor 2015; Ivani Rossi 2010; Maia et al. 2014; Universidade do Café 2025), only 8% reported this habit in the present study. This reduction may reflect growing awareness of the beverage's stimulating effects. Supporting this hypothesis, earlier studies found that nearly 30% of parents with children under 9 years old avoided offering them coffee, citing concerns about its impact on sleep (Ivani Rossi 2010). Similarly, in the present survey, the most frequently cited adverse effects of coffee ingestion were insomnia, dental damage, heartburn, anxiety, and gastritis, suggesting that late consumption may be avoided as a precautionary measure.

The present study also demonstrated that coffee consumption in Brazil continues to occur predominantly at home (95%), followed by the homes of friends and family (55%), the workplace (60%), coffee shops (59%), bakeries (36%), restaurants (31%), and educational institutions (26%) (Table 6). These statistics reaffirm the central role of the domestic environment in shaping consumption habits while also highlighting the relevance of social and professional contexts.

Data from previous studies provide historical context for these patterns. A survey conducted by ABIC, which compared data from 2003 to 2010, reported a 300% increase in out-of-home coffee consumption over that period, with bakeries cited as the most common location outside the home (Ivani Rossi 2010). Subsequent investigations confirmed that this behavior was more frequent among men and individuals from higher-income households, reflecting gender and socioeconomic disparities in access to out-of-home coffee consumption (Maia et al. 2014; Pino et al. 2017). Moreover, the Euromonitor consulting (2015) survey confirmed a continued rise in out-of-home consumption,

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TABLE 6 | Participant characteristics: age of start, consumption frequency, location, time of day, and reasons for coffee consumption, overall and by group (sweetened coffee [SC] and unsweetened coffee [UC] consumers).

		Total		SC		UC	
Question	**Response/answer**[a]	**1200**	**100%**	**550**	**46%**	**650**	**54%**
How often do you consume coffee?	Daily (two or more times a day)	73%		372	68%^{B}	506	78%^{A}
	Daily (one time a day)	18%		114	21%^{A}	99	15%^{B}
	1–3 times a week	5%		33	6%^{A}	29	4%^{A}
	Occasionally	4%		30	5%^{A}	19	3%^{B}
	Others	0%		1	0%^{A}	1	0%^{A}
What time of day do you usually consume coffee?	At breakfast	82%		455	83%^{A}	528	81%^{A}
	Between breakfast and lunch	37%		170	31%^{B}	279	43%^{A}
	After lunch	41%		173	31%^{B}	318	49%^{A}
	Afternoon snack	47%		253	46%^{A}	314	48%^{A}
	After dinner	8%		58	11%^{A}	41	6%^{B}
What is the reason for drinking coffee?	Product flavor	75%		377	69%^{B}	518	80%^{A}
	Product aroma	56%		256	47%^{B}	410	63%^{A}
	Health benefits	16%		60	11%^{B}	134	21%^{A}
	Improves disposition	55%		263	48%^{B}	392	60%^{A}
	Improves mood	28%		128	23%^{B}	212	33%^{A}
	Pleasure	58%		272	49%^{B}	419	64%^{A}
	Socialization	31%		135	25%^{B}	240	37%^{A}
	Habit	61%		334	61%^{A}	394	61%^{A}
	Others	1%		12	2%^{A}	0	0%^{B}
How old were you when you started drinking coffee? (approximate age)	Under 18 years old	71%		419	76%^{A}	429	66%^{B}
	18–24 years old	15%		68	12%^{B}	113	17%^{A}
	25–39 years old	8%		32	6%^{B}	58	9%^{A}
	40–55 years old	1%		2	0%^{B}	13	2%^{A}
	Over 55 years old	0%		0	0%	0	0%
	Other (I do not know)	6%		29	5%^{A}	37	6%^{A}
Where did your coffee consumption begin?	Home	85%		486	88%^{A}	537	83%^{B}
	Work	17%		72	13%^{B}	126	19%^{A}
	House of friends and family	9%		47	9%^{A}	65	10%^{A}
	Coffee shop	4%		11	2%^{B}	36	6%^{A}
	Restaurant	1%		6	1%^{A}	8	1%^{A}
	Bakery	2%		9	2%^{A}	17	3%^{A}
	Educational environment	7%		29	5%^{A}	49	8%^{A}
	Others	0%		3	1%^{A}	0	0%^{B}

(Continues)

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TABLE 6 | (Continued)

Question	Response/answer[a]	Total		SC		UC	
		1200	100%	550	46%	650	54%
Where do you currently consume coffee?	Home	95%		517	94%^{A}	624	96%^{A}
	Coffee shop	59%		266	48%^{B}	438	67%^{A}
	Restaurant	31%		152	28%^{B}	223	34%^{A}
	House of friends and family	55%		291	53%^{A}	372	57%^{A}
	Work	60%		296	54%^{B}	428	66%^{A}
	Bakery	36%		180	33%^{B}	247	38%^{A}
	Educational environment	26%		141	26%^{A}	175	27%^{A}
	Others	1%		6	1%^{A}	0	0%^{B}
Your coffee consumption during the COVID-19 pandemic:	Increased	29%		144	26%^{B}	207	32%^{A}
	Did not change	63%		360	65%^{A}	394	61%^{B}
	Decreased	8%		46	8%^{A}	49	8%^{A}
Which of the following items do you consider to be a negative aspect of coffee consumption?	Insomnia	33%		189	34%^{A}	207	32%^{A}
	Anxiety	25%		131	24%^{A}	171	26%^{A}
	Heartburn	27%		158	29%^{A}	164	25%^{A}
	Gastritis	24%		153	28%^{A}	134	21%^{B}
	Nausea	4%		29	5%^{A}	16	2%^{B}
	Intestinal discomfort	13%		85	15%^{A}	65	10%^{B}
	Headache (if you don't drink coffee)	30%		164	30%^{A}	191	29%^{A}
	Addiction	20%		121	22%^{A}	124	19%^{A}
	Tooth damage	30%		157	29%^{A}	208	32%^{A}
	Increases the risk of cardiovascular disease	3%		21	4%^{A}	20	3%^{A}
	Reduces iron absorption	9%		49	9%^{A}	64	10%^{A}
	None of the above	11%		61	11%^{A}	74	11%^{A}
Are you aware of the health benefits of drinking coffee?	Yes	70%			74%^{A}	—	65%^{B}
	No	30%			26%^{B}	—	35%^{A}

Note: Different capital letters in the same line indicate statistical difference ($p \leq 0.10$) by the k-proportions test between SC and UC.
[a]Expressed in absolute numbers and/or percentages.

identifying coffee shops as the second most cited venue, after workplaces and bakeries. Other national dietary data also indicate a continued rise in the frequency of going out to coffee shops, suggesting an ongoing diversification of consumption environments and social contexts (Bezerra et al. 2017). This trend is consistent with the results of the present study, which reveal the growing relevance of coffee shops and other out-of-home venues within Brazilian coffee culture.

More recent publication continues to identify home and workplace as the primary locations for coffee consumption. In 2021 the home predominated as the main place of coffee consumption—likely due to pandemic restrictions—while in 2023 and 2025, the workplace had regained the top position, relegating home consumption to second place, and cafés, bars, and restaurants appearing in third place (Axxus 2025). The same survey reports that cafeteria visits dropped to about 9% in 2021 but rebounded to around 51% in 2023 (Axxus 2019, 2021, 2023).

Similar results were found in the present study, with 59% of participants reporting coffee consumption in coffee shops (Table 6). Although this figure is slightly higher than those previously noted, it is likely influenced by the socioeconomic profile of the sample, which included a greater proportion of individuals with higher income and education levels. Still, the consistency across studies suggests that the increasing popularity of coffee shops as spaces for consumption is a broader and ongoing trend in Brazilian coffee culture.

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The resilience of coffee consumption among Brazilians was further demonstrated during the COVID-19 pandemic: in the present investigation, 63% of participants reported no change in consumption volume, while 29% reported an increase. These results are consistent with previous surveys, which have linked this tendency to the increased time spent at home (Axxus 2021; Soares, Crepalde, et al. 2023). Similar behavior had already been observed during the economic crises of 2014–2016, when rising prices and reduced household income did not affect overall consumption volume, but instead shifted consumer behavior, favoring lower-cost brands and reducing out-of-home consumption. Studies conducted in 2014/2015 reported that Brazilians prepared coffee beverages, on average, twice a day (Euromonitor 2015, 2016; Maia et al. 2014). In comparison, 75% of respondents in 2023 stated that they brewed coffee more than twice daily (Axxus 2023), during a period of economic improvement in the country. Collectively, these data underscore that while coffee consumption in Brazil remains remarkably stable, improvements in financial conditions tend to encourage increased frequency and diversification of consumption habits.

When it comes to why Brazilians drink coffee, flavor takes the lead. For many participants, coffee is not merely a source of energy but a pleasurable sensory experience. Seventy-five percent cited flavor as their main motivation, followed closely by habit (61%), pleasure (58%), and aroma (56%) (Table 6). Other reasons were also mentioned, including enhancing disposition (55%), promoting social interaction (31%), improving mood (28%), and even health-related benefits (16%).

Earlier studies identified flavor and aroma as the main drivers of at-home coffee consumption among Brazilian adults (Euromonitor 2015; Sousa et al. 2016). More recent investigations, however, indicate a broader set of motivations that encompass both functional and emotional dimensions. Across different populations, coffee has been associated with positive feelings such as energy, satisfaction, and pleasure, as well as with social and cultural meanings (Samoggia and Riedel 2019; Samoggia et al. 2020). During the COVID-19 pandemic, these emotional factors became even more evident: in a Portuguese sample, participants reported increasing coffee intake to relieve stress, maintain alertness, and reinforce social and cultural habits (Batista et al. 2023). In 2023, a Brazilian survey (Axxus 2023) described that mood and disposition were the most frequently mentioned reasons, followed by ritual, pleasure, well-being, and social interaction—highlighting how deeply coffee is embedded in both the routine and emotional fabric of Brazilian life.

Beyond sensory appeal and emotional connections, coffee also has a functional dimension that consumers are increasingly recognizing. Seventy percent of participants reported being aware of the health-related benefits associated with coffee consumption (Table 6), most notably its stimulant (63%) and antioxidant (8%) properties. This awareness aligns with previous results. In a 2010 survey, for instance, 50% of respondents indicated some familiarity with the positive health effects of coffee, although many younger and lower-income individuals were unable to specify them. Among those who could, improvements in mood and memory were commonly cited, with mentions of disease prevention and antioxidant content appearing less frequently (Ivani Rossi 2010).

In a 2014 study, consumers exhibited similar behavior regarding knowledge about health benefits (Maia et al. 2014); however, this topic was not addressed in more recent surveys. Our results are similar to those reported by Corso et al. (2018), when they investigated the role of sociodemographic, cognitive, and behavioral characteristics in the acceptance of functional foods by Brazilian coffee consumers, in which 50% of the participants had heard of the health benefits of coffee consumption, the most cited being the stimulating (49%) and antioxidant (26%) action. In both consulted studies, the effects of the coffee beverage as a stimulant and antioxidant were highlighted, suggesting that Brazilian consumers' perception of coffee's health benefits has remained relatively consistent over time.

While the beneficial effects of coffee are well recognized, participants also acknowledged potential drawbacks associated with its consumption. The most frequently reported adverse consequences included gastric discomfort—such as heartburn, gastritis, and nausea—cited by 55% of respondents, followed by insomnia (33%), dental damage (30%), anxiety (25%), and intestinal discomfort (13%) (Table 6). Additionally, a significant number of participants mentioned symptoms linked to withdrawal or dependency, including headaches when abstaining (30%) and a perceived sense of addiction (20%). Research by Arruda et al. (2009), Maia et al. (2014), and Violin et al. (2020) similarly identified sleep disturbances, gastrointestinal effects, and perceived dependency as common concerns. As previously discussed, the concern about insomnia may also help explain the relatively low rate of evening coffee consumption observed in the current investigation (Table 6), suggesting that some individuals adjust their intake to mitigate such effects.

Although certain medical conditions may require restricted coffee intake, the overall evidence strongly supports its health benefits for healthy adults, a conclusion consistently backed by the scientific literature (Makiso et al. 2024; Safe et al. 2023; Simon et al. 2022). Nonetheless, in a country where coffee is deeply ingrained in daily life, a considerable share of consumers still lacks clear information about its positive attributes. This is noteworthy even within the present study's sample, which was composed predominantly of individuals with higher income and education levels than the national average.

Such gaps in knowledge underscore the importance of enhancing public access to accurate, evidence-based information on the health effects of coffee consumption. As early as 2014, it was reported that many consumers refrained from drinking coffee due to misconceptions about its harmful effects (Maia et al. 2014). These persistent beliefs highlight the need for targeted educational efforts, a topic that will be explored in more detail later in this work.

3.3 | Coffee Preparation, Consumption Preferences, and Purchasing Behavior

This section aims to address how the beverage is typically prepared and consumed, as well as assess consumer preferences

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and purchasing behaviors, providing useful insights into future cultural, economic, and technological trends.

Among the participants, the consumption of pure coffee was predominant (80%), as was the use of filtered coffee prepared with a cloth strainer or paper filter—cited as both the most frequent brewing method at home (73%) and the preferred one (75%) (Table 7). Espresso was reported as the second favorite type of coffee beverage (37%), showing substantial preparation rates at home, either using an espresso machine (37%) or single-dose/capsule systems (28%), and standing out as the most frequently consumed type in coffee shops (61%), almost twice the rate of filtered coffee (33%) (Table 7).

Filtered coffee remains the most traditional and established method of home preparation. However, notable shifts have occurred over time: instant coffee has experienced a 2.4-fold decrease in consumption, while espresso beverages (espresso machine and capsule) have seen a more than fivefold increase in popularity (Axxus 2023; Ivani Rossi 2010).

The global espresso market was valued at USD 4.12 billion in 2024 (Market Data Forecast 2025). It is projected to reach USD 8.05 billion by 2033, registering a compound annual growth rate (CAGR) of 7.72% over the forecast period (2025–2033). In Brazil, the popularity of espresso coffee has increased significantly in recent decades. Once primarily associated with out-of-home settings, espresso gradually expanded into the domestic sphere (Ivani Rossi 2010; Nielsen 2013). This transition was largely facilitated by the introduction of capsule systems, which offered consumers greater convenience and access to a wide range of blends and flavours (Euromonitor 2016, 2017). The breaking of Nespresso's patent in the mid-2010s further accelerated this movement, enabling the proliferation of compatible machines and capsules from various brands and making espresso preparation more accessible (Conley et al. 2013; Euromonitor 2015).

One particularly noteworthy finding of this study was the high proportion of participants who reported not adding a sweetener to their coffee (54%) (Table 7). A similar trend was observed in a recent study involving 300 coffee consumers across five regions of Brazil, in which 51% of respondents indicated that they do not add sweetener to their coffee (Universidade do Café 2025).

These results stand in contrast to long-standing patterns observed in national dietary surveys, which have historically described a strong tendency among Brazilians to add sugar or sweeteners—not only to coffee, but also to other commonly consumed beverages such as juice, milk-based drinks, and tea. For instance, an analysis based on the 2008–2009 National Dietary Survey found that only 1% of respondents ($n = 34{,}003$, aged 10 years or older) consumed coffee without any form of sweetener (Sousa and Costa 2015). A more recent study, using data from the 2017–2018 National Household Budget Survey, revealed a modest shift: although the proportion of consumers who refrained from sweetening had increased, 88% of participants ($n = 28{,}153$, aged 20–59) still reported adding sugar to their coffee (Ued et al. 2024).

These contrasting figures suggest a potential behavioral transition underway among Brazilian coffee consumers, a trend that gains relevance considering growing public health concerns related to sugar consumption in the country and globally. Given this context, the differences in behaviors, motivations, and profiles of SC and UC coffee consumers identified in the present study will be explored in greater detail in the next section.

When contextualized with data from previous studies (Axxus 2023; Euromonitor 2015, 2016, 2017; Maia et al. 2014; Nielsen 2013; SPCH-ABIC 2021), the results of this research reveal an evolution in coffee consumption preferences and habits in Brazil. Traditional coffee remains the most frequently consumed variety (61%), followed by gourmet and superior categories (44%), which are predominantly purchased in roasted and ground form (83%) (Table 7). This pattern reflects a trajectory already observed in earlier surveys, yet it also highlights a significant shift in consumer preferences over the past decade.

Until 2015, concerns about coffee quality were rarely emphasized in consumer surveys, and gourmet options had limited appeal among consumers. However, in subsequent years, the increased availability of higher-quality coffees, with enhanced sensory attributes, led to a rapid expansion of this market segment. By the beginning of this decade, a shift in consumption patterns was detected: the share of superior, specialty, and gourmet coffees (38%) surpassed that of traditional coffee (32%) (Axxus 2025), signaling a growing appreciation for differentiated products and a maturing consumer profile.

In terms of purchase locations, markets and supermarkets remain the primary choice for most consumers (90%) (Table 7)—a pattern consistently reported in previous consumer surveys (Axxus 2023, 2025; Euromonitor 2017; Maia et al. 2014; Nielsen 2013). However, the present study reveals an important increase in purchases made through online platforms (19%) and coffee shops (11%) (Table 7), exposing a gradual diversification of purchase channels.

Earlier studies had already identified an expansion in the points of sale following the introduction of coffees with improved sensory quality. At that time, coffee shops/cafeterias accounted for only 3% of purchases, while online stores and marketplaces represented 2%, and coffee clubs a mere 0.5% (Axxus 2023; Euromonitor 2017; Maia et al. 2014; Nielsen 2013).

Considering purchasing criteria, brand (68%) and price (64%) remain the most influential factors for consumers when choosing coffee products (Table 7), as also shown in previous research (Axxus 2023; Costa et al. 2024; Euromonitor 2017; Ivani Rossi 2010; Maia et al. 2014; Nielsen 2013). Interestingly, the literature suggests that while economic fluctuations may influence brand selection, prompting consumers to opt for lower-cost options within their preferred brand set, such changes do not necessarily alter overall consumption habits (Baylan 2017; Euromonitor 2015). This behavior highlights the centrality of coffee in Brazilian daily life, where brand loyalty can coexist with price sensitivity, particularly during periods of financial constraint.

Forty-three percent of respondents indicated in the present study that the presence of a quality seal influences their purchasing decisions, and it was also the most frequently mentioned attribute

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justifying a willingness to pay a premium (53%) (Table 7). This consumer behavior aligns with the increase in coffee certification by ABIC in the last years, as well as with the broader recognition of the organization's efforts to enhance product quality over the past three decades (ABIC 2025). Through the certification of brands based on cup quality and purity standards, ABIC seeks to promote transparency and elevate industry practices. The growing consumer interest in certification labels may thus

TABLE 7 | Coffee consumption habits: drinking, brewing, and purchasing patterns, overall and by group (sweetened coffee [SC] and unsweetened coffee [UC] consumers).

		Total		SC		UC	
Question	**Response/answer[a]**	**1200**	**100%**	**550**	**46%**	**650**	**54%**
How do you usually drink your coffee?	Pure	80%		380	69%^{B}	580	89%^{A}
	With milk	21%		168	31%^{A}	82	13%^{B}
	Others	1%		15	3%^{A}	2	0%^{B}
Which coffee brewing method do you use at home?[2]	Drip (cloth strainer or paper filter)	73%		400	73%^{A}	478	74%^{A}
	Drip coffee maker	24%		115	21%^{B}	178	27%^{A}
	Espresso machine	37%		149	27%^{B}	296	46%^{A}
	Espresso machine—capsule	28%		115	21%^{B}	226	35%^{A}
	French press	14%		38	7%^{B}	135	21%^{A}
	Italian or Moka	23%		105	19%^{B}	175	27%^{A}
	Chemex	2%		9	2%^{A}	19	3%^{A}
	Hario V60	7%		13	2%^{B}	73	11%^{A}
	Turkish	1%		1	0%^{A}	5	1%^{A}
	Siphon	0%		1	0%^{A}	2	0%^{A}
	Instant	21%		127	23%^{A}	121	19%^{B}
Which is your favorite coffee brewing method?[b]	Drip (cloth strainer or paper filter)	75%		443	81%^{A}	452	70%^{B}
	Espresso	37%		168	31%^{B}	280	43%^{A}
	French press	13%		39	7%^{B}	112	17%^{A}
	Italian or Moka	18%		76	14%^{B}	138	21%^{A}
	Chemex	2%		3	1%^{B}	23	4%^{A}
	Hario V60	7%		5	1%^{B}	80	12%^{A}
	Turkish	1%		6	1%^{A}	4	1%^{A}
	Siphon	0%		1	0%^{A}	2	0%^{A}
	Instant	8%		57	10%^{A}	42	6%^{B}
Which type of coffee do you use at home?	Traditional	61%		404	73%^{A}	331	51%^{B}
	Extra-strong	22%		132	24%^{A}	133	20%^{A}
	Gourmet and superior	44%		161	29%^{B}	372	57%^{A}
	Other	1%		3	1%^{A}	3	1%^{A}
In what form do you usually buy coffee?	Roasted and ground	83%		474	86%^{A}	524	81%^{B}
	Roasted bean	19%		47	9%^{B}	181	28%^{A}
	Instant	20%		128	23%^{A}	115	18%^{B}
	Capsule	23%		98	18%^{B}	177	27%^{A}

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TABLE 7 | (Continued)

Question	Response/answer[a]	Total		SC		UC	
		1200	100%	550	46%	650	54%
Where do you or your family usually buy coffee for home use?	Market/supermarket	90%		533	97%^{A}	549	84%^{B}
	Coffee shop	11%		28	5%^{B}	101	16%^{A}
	Online	19%		54	10%^{B}	176	27%^{A}
	Producer	4%		6	1%^{B}	40	6%^{A}
What do you take into consideration at the time of purchase?	Price	64%		361	66%^{A}	406	62%^{A}
	Brand	68%		416	76%^{A}	397	61%^{B}
	Roasting degree	44%		166	30%^{B}	367	56%^{A}
	Information on the label about coffee category	49%		204	37%^{B}	387	60%^{A}
	Quality and Purity seals	43%		191	35%^{B}	328	50%^{A}
	Producing region	27%		93	17%^{B}	227	35%^{A}
	Origin indication	26%		108	20%^{B}	204	31%^{A}
	Sustainability information	17%		61	11%^{B}	137	21%^{A}
	Packaging type and material	13%		61	11%^{B}	99	15%^{A}
	Best before date	33%		207	38%^{A}	186	29%^{B}
	Availability at the sales point	17%		97	18%^{A}	112	17%^{A}
Would you be willing to pay more for a coffee with any of the characteristics mentioned above?	Yes	82%		434	79%^{B}	554	85%^{A}
	No	18%		116	21%^{A}	96	15%^{B}
If yes, please specify which characteristics	Brand	27%		171	31%^{A}	153	24%^{B}
	Roasting degree	41%		184	33%^{B}	308	47%^{A}
	Information on the label about coffee category	42%		203	37%^{B}	299	46%^{A}
	Quality seals	53%		263	48%^{B}	370	57%^{A}
	Producing region	33%		154	28%^{B}	236	36%^{A}
	Origin indication	34%		166	30%^{B}	242	37%^{A}
	Sustainability information	28%		147	27%^{A}	191	29%^{A}
	Packaging type and material	16%		87	16%^{A}	101	16%^{A}
	Best before date	13%		91	17%^{A}	62	10%^{B}
Which coffee species or coffee varieties do you know?	Arabica	78%		369	67%^{B}	572	88%^{A}
	Canephora	13%		40	7%^{B}	121	19%^{A}
	Conilon	36%		147	27%^{B}	279	43%^{A}
	Robusta	36%		148	27%^{B}	289	44%^{A}
	I don't know	21%		174	32%^{A}	83	13%^{B}

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TABLE 7 | (Continued)

Question	Response/answer[a]	Total		SC		UC	
		1200	100%	550	46%	650	54%
Do you know which coffee you usually consume in terms of species and variety?	Arabica	56%		243	44%^{B}	434	67%^{A}
	Canephora	2%		4	1%^{B}	18	3%^{A}
	Conilon	6%		27	5%^{A}	42	6%^{A}
	Robusta	5%		24	4%^{A}	40	6%^{A}
	I don't know	34%		250	45%^{A}	153	24%^{B}
Which roasting degree do you prefer?	Light	12%		40	7%^{B}	109	17%^{A}
	Medium	63%		295	54%^{B}	457	70%^{A}
	Dark	22%		149	27%^{A}	113	17%^{B}
	I don't know	19%		125	23%^{A}	97	15%^{B}
How often do you go to coffee shops?	Weekly	16%		65	12%^{B}	121	19%^{A}
	Monthly	15%		66	12%^{B}	115	18%^{A}
	Occasional	54%		296	54%^{A}	352	54%^{A}
	I don't go often	16%		123	22%^{A}	66	10%^{B}
What type of coffee do you usually order when visiting coffee shops?[b]	Drip	33%		151	27%^{B}	242	37%^{A}
	Espresso	61%		294	53%^{B}	440	68%^{A}
	French press	11%		31	6%^{A}	99	15%^{A}
	Italian or Moka	11%		48	9%^{B}	84	13%^{A}
	Chemex	3%		4	1%^{B}	35	5%^{A}
	Hario V60	6%		4	1%^{B}	69	11%^{A}
	Turkish	1%		6	1%^{A}	8	1%^{A}
	Siphon	0%		1	0%^{A}	4	1%^{A}
	Coffee-based preparations such as cappuccino and coffee with milk	37%		215	39%^{A}	230	35%^{A}
	I do not go to coffee shops	11%		85	15%^{A}	46	7%^{B}

Note: Different capital letters in the same line indicate statistical difference ($p \leq 0.10$) by the k-proportions test between SC and UC.
[a]Expressed in absolute numbers and/or percentages.
[b]Pictures of brewing equipment were presented in the questionnaire to help consumers identify each method.

be interpreted as a direct outcome of these initiatives, reflecting increased trust in the official guarantees displayed on coffee packaging.

Coffee quality perception encompasses both intrinsic and extrinsic factors, such as aroma, flavor, body, roasting degree, brand, and certification. Recent approaches have expanded the concept of coffee quality to encompass not only sensory and chemical aspects but also cultural and microbiological dimensions that affect consumer perception (Pimenta et al. 2018). These multidimensional influences highlight how consumers integrate objective and subjective criteria when defining a "quality coffee," linking sensory appreciation with emotional, cultural, and informational cues. In the present study, when participants were asked to mention the first four words that came to mind when thinking of quality coffee, most responses referred to sensory attributes (86%) and general notions of quality (56%) (Table 8). These perceptions align with the main reasons mentioned for coffee consumption, primarily flavor (75%) and aroma (56%) (Table 6), as well as with the attributes considered at the time of purchase, beyond price and brand (Table 7).

Among the product attributes most closely linked to sensory perception, roasting degree stood out in this study as a relevant factor. At the time of purchase, nearly half of the participants (44%) considered it, and it was ranked as the second highest-mentioned feature for which consumers were willing to pay more (41%) (Table 7). This result is consistent with earlier observations reported by Euromonitor Consulting (2017), which noted that the Brazilian market was progressing through successive waves of coffee consumption. According to the report, consumers frequenting coffee shops were already demonstrating greater curiosity about the origin of the beans, their sensory profiles, and the types of roasting.

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TABLE 8 | Relative importance of the categories identified when participants were asked to cite the first four words they associated with a quality coffee.

Category	Participants who cited the category[a]	Total citations in the category[b]
Sensory		
Terms related to the sensory characteristics of the coffee beverage.	1035 (86%)	2340 (56%)
Processing and quality		
Considers the coffee origin and genetics, cultivation practices, processing stages, and terms associated with quality.	677 (56%)	947 (23%)
Feelings/memories		
Included evoked feelings or memories.	116 (10%)	131 (3%)
Brewing		
This deals with characteristics and preparation methods.	151 (13%)	173 (4%)
Health		
It addressed the positive and negative aspects of coffee consumption on health.	247 (21%)	334 (8%)
Moment, habit, and socialization		
This is related to the moment, period, and customs of consumption, food, and social interactions.	97 (8%)	110 (3%)
Price/availability		
It included the amount charged, producer appreciation, and market availability.	132 (11%)	154 (4%)

[a]Total number of participants who mentioned the category (absolute number, percentage).
[b]Total number of citations in the category (absolute number, percentage).

In the current investigation, medium-roast coffee emerged as the preferred option for most participants (63%), indicating a reduced preference for darker roasting degrees, which were favored by only 22% (Table 7). This preference mirrors the proportion of consumers who reported purchasing extra-strong coffee (22%) (Table 7), a category typically associated with darker roast levels. This trend is consistent with recent research on coffee consumers in Brazil (Universidade do Café 2025).

While dark roasts have historically dominated Brazilian consumption patterns (Monteiro et al. 2010), the growing appreciation for flavor complexity and bean diversity appears to be shifting preferences toward lighter roasting profiles, such as medium roasts, which allow for the expression of each coffee's intrinsic qualities, resulting in more balanced and nuanced sensory experiences (Schenker and Rothgeb 2017).

In terms of knowledge and consumption of different coffee species, *Coffea arabica* was the most recognized and widely consumed among survey participants: 78% reported familiarity with the species, and 56% indicated that they consumed it (Table 7). Although *Coffea canephora*, including its Conilon and Robusta varieties, was also mentioned, its reported consumption was considerably lower (Table 7). This result is consistent with Brazil's historical emphasis on research and development focused on *C. arabica*, as well as the fact that, until recently, *C. canephora* was predominantly used in soluble coffee production or blended with *C. arabica* in roasted and ground products (Fernandes et al. 2024).

However, the last decade has seen a significant change. Increased investment in *C. canephora* research has led to advancements in both agronomic traits, particularly productivity, and cup quality (Fernandes et al. 2024; Viencz et al. 2023). Known as "fine Robustas", high-quality *C. canephora* coffees have gained recognition from the Specialty Coffee Association as having potential for the specialty segment (SCA 2020). This recognition is already reflected in the Brazilian market, where two commercial brands, Melitta and Três Corações, have launched 100% *C. canephora* products (Melitta 2025; Três Corações 2025), helping to expand consumer exposure and acceptance of this species.

Taken together, the results shown in this section depict a coffee consumer who is increasingly engaged, informed, and attentive to quality. Interest in preparation methods, origin, roasting degree process, and the presence of a quality seal—along with a growing willingness to pay more for perceived value—highlights a shift toward more conscious consumption. Notably, the substantial presence of enthusiasts outside the production chain suggests that appreciation for sensory experience and coffee culture is expanding among the general public. Building on this profile, the following section explores

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how preferences differ between consumers of SC and UC coffee beverages.

3.4 | Comparing the Behavior of Consumers of SC and UC Beverages

To deepen the understanding of consumer profiles, participants were divided into two groups based on their consumption of SC or UC coffee beverages, with the groups being relatively balanced: 46% belonged to the SC group and 54% to the UC group (Table 9). Among SC consumers, table sugar (sucrose) was by far the most used sweetener, cited by 78% of respondents. Reported amounts varied considerably: 42% added up to 3 g per 100 mL cup, 23% used between 3 and 6 g, and 14% declared adding quantities between 6 and 24 g.

The primary reasons reported for sweetening included the bitterness of the beverage (46%), habitual behavior (42%), and a general preference for sweet drinks (27%) (Table 9). Despite these motivations, over half of the SC group (53%) indicated a reduction in the amount of sugar used in recent years. They reported that it was mainly due to health concerns (35%) and dietary re-education (22%), but also cited other reasons, such as changes in taste (22%), perceived improvements in coffee quality (17%), and enhanced flavor of the brew itself (14%) (Table 9).

The addition of large amounts of sugar is traditional among Brazilian coffee drinkers. Based on consumption data from 2008/2009, Souza et al. (2013) estimated that sugar added to the coffee brews represented approximately 10% of the beverage volume when only sugar was used, and around 5% when sugar was combined with artificial sweeteners. Sensory studies from the same period indicated that the ideal sweetness level for roasted coffee brews among Brazilian consumers ranged between 12% and 12.5% sugar (Fonteles et al. 2010; Moraes and Bolini 2010). More recently, Viencz et al. (2024) reported, in a home-use test, that individuals who SC their beverage typically used between 1/4 and 1 packet (5 g) of sugar per 100 mL. Similarly, Portela et al. (2024), in a study comparing the behavior of SC and UC coffee drinkers, found that 60% of SC consumers added up to 6 g of sugar per 100 mL of beverage, and the majority (53%) stated they had been reducing the amount of sugar used.

Despite the variability in reported quantities, these evaluations suggest a downward trend in sugar addition compared to the previous decade. This aligns with the present study's results and indicates that, alongside a growing number of consumers who no longer sweeten their coffee, those who continue to do so are progressively reducing the amount added.

Another interesting point to consider is that 28% of SC consumers reported using sweeteners instead of sucrose to achieve the desired sweetness on their coffee drinks (Table 9). Although the reasons or timing of this substitution were not explored in the survey, this behavior reinforces the broader trend toward reducing sugar intake. Even among those who continue to prefer SC beverages, there appears to be an active effort to reduce the use of added sugars.

In recent research, stevia was reported as the most commonly used sweetener among SC consumers, confirming the trend among Brazilians to consume coffee in a healthier way (Universidade do Café 2025).

The decision to reduce or eliminate added sugar in the coffee brew appears to be related to changes in the behavioral profile of consumers. This interpretation is supported by the fact that 76% of UC participants reported having, in the past, the habit of sweetening their coffee. Among them, 38% had either stopped more than a decade ago or had never added sugar. In comparison, the majority (39%) reported discontinuing the habit within the last 5 years (Table 9), indicating that this is a relatively recent trend. Perhaps for this reason, it is less captured in official consumer surveys, especially considering that ABIC surveys conducted over the past decade have not addressed this topic.

The reasons cited by UC participants for avoiding sugar were diverse but notably aligned with the motivations of SC consumers who reported reducing their sugar consumption. Key factors included appreciation for the flavor of the coffee brew (62%), perceived improvements in coffee quality (46%), changes in individual taste preferences (45%), as well as health (45%) and dietary (28%) concerns, and habit (22%) (Table 9). Together, these data reinforce the observation that Brazilian consumers are moving toward more conscious and selective coffee consumption practices.

Portela et al. (2024), in a study with Brazilian consumers of SC and UC coffee beverages, highlighted that UC participants reported having stopped adding or never added sugar to the coffee beverage for reasons related to the flavor of the coffee brew (48%), diet and health concerns (41%), improvement in coffee quality (30%), and changes in their taste (30%). These results align with the findings of the present study.

A recent report, based on data from the 2017–2018 National Household Budget Survey, examined sucrose intake and its dietary sources among Brazilian adults. The authors indicated that the main way sugar was introduced into diet in homemade preparations was via beverages such as coffee (56%), juice (34%), milk-based preparations and smoothies (3%), powdered or processed juice (3%), whole milk (2%), and tea (2%) (Ued et al. 2024). Other research has similarly shown that high sugar intake among the Brazilian population is already reflected in the growing prevalence of weight-related diseases (Brasil 2024; Siqueira et al. 2021).

This consumption pattern underscores the need for continuous investigation, given its potential implications for public health and its significance to both the food industry and the coffee service sector. Evidence from this study may guide coffee shops and manufacturers to reconsider product formulations to align with healthier consumption trends. Gradual sugar reduction strategies, such as those proposed by Portela et al. (2024), which define acceptable sensory thresholds to minimize impact on consumer acceptance, offer a promising pathway. Furthermore, the incorporation of alternative sweetening agents represents an additional strategy to simultaneously promote consumer health and foster product innovation over the long term.

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TABLE 9 | Information on reasons and habits for consuming sweetened (SC) and unsweetened (UC) coffee beverages, and neophobia behavior.

		Total		SC		UC	
Question	**Response/answer[a]**	**1200**	**100%**	**550**	**46%**	**650**	**54%**
Why do you sweeten your coffee beverage?	Bitterness	—	—	255	46%	—	—
	I like sweet sdrinks	—	—	147	27%	—	—
	Habit	—	—	232	42%	—	—
	Other:	—	—	33	6%	—	—
What do you use to sweeten it with?	Sugar	—	—	427	78%	—	—
	Sweetener	—	—	156	28%	—	—
	Other	—	—	0	0%	—	—
How much sugar/sweetener do you add to your coffee beverage (in a 100 mL cup)?[b]	Up to 3 g of sugar	—	—	230	42%	—	—
	3 g to 6 g of sugar	—	—	128	23%	—	—
	6 g to 12 g of sugar	—	—	49	9%	—	—
	12 g to 24 g of sugar	—	—	30	5%	—	—
	Up to 5 drops of sweetener	—	—	81	15%	—	—
	6 drops to 10 drops of sweetener	—	—	5	1%	—	—
	Over 10 drops of sweetener	—	—	11	2%	—	—
	Other	—	—	14	2%	—	—
Regarding the amount of sugar you add to your coffee, would you say that in the last few years it has	Decrease	—	—	46	8%	—	—
	Did not change	—	—	201	36%	—	—
	Increase	—	—	294	53%	—	—
Whether your sugar consumption has decreased, what was the reason?	Health concerns	—	—	195	35%	—	—
	Diet (dietary re-education)	—	—	122	22%	—	—
	Flavor of the beverage	—	—	74	13%	—	—
	Improves in coffee quality	—	—	94	17%	—	—
	Change of taste	—	—	123	22%	—	—
	Others	—	—	4	1%	—	—
Did you have the habit of adding sugar to your coffee?	Yes	—	—	—	—	495	76%
	No	—	—	—	—	155	24%
If YES, how long has it been since you added sugar to your coffee?	Up to 1 year	—	—	—	—	55	8%
	Between 1 and 5 years	—	—	—	—	200	31%
	Between 5 and 10 years	—	—	—	—	150	23%
	Over 10 years or never sweetened	—	—	—	—	245	38%
Why did you stop/not add sugar to your coffee?	Flavor of the beverage	—	—	—	—	404	62%
	Improvementin coffee quality	—	—	—	—	299	46%
	Health concerns	—	—	—	—	291	45%
	Diet (dietary re-education)		—	—	—	180	28%
	Change of taste	—	—	—	—	290	45%
	Habit	—	—	—	—	144	22%

(Continues)

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TABLE 9 | (Continued)

Question	Response/answer[a]	Total 1200	Total 100%	SC 550	SC 46%	UC 650	UC 54%
Neophobia	Neutral	66%		366	67%[A]	425	65%[A]
	Neophilic	17%		67	12%[B]	142	22%[A]
	Neophobic	17%		117	21%[A]	83	13%[B]

Note: Different capital letters in the same line indicate statistical differences ($p \leq 0.10$) by the k-proportions test between SC and UC.
[a]Expressed in absolute numbers and/or percentages.
[b]Household measurement figures, with different types of spoons and sachets, were presented in the questionnaire to guide consumers in estimating the amount of sugar.

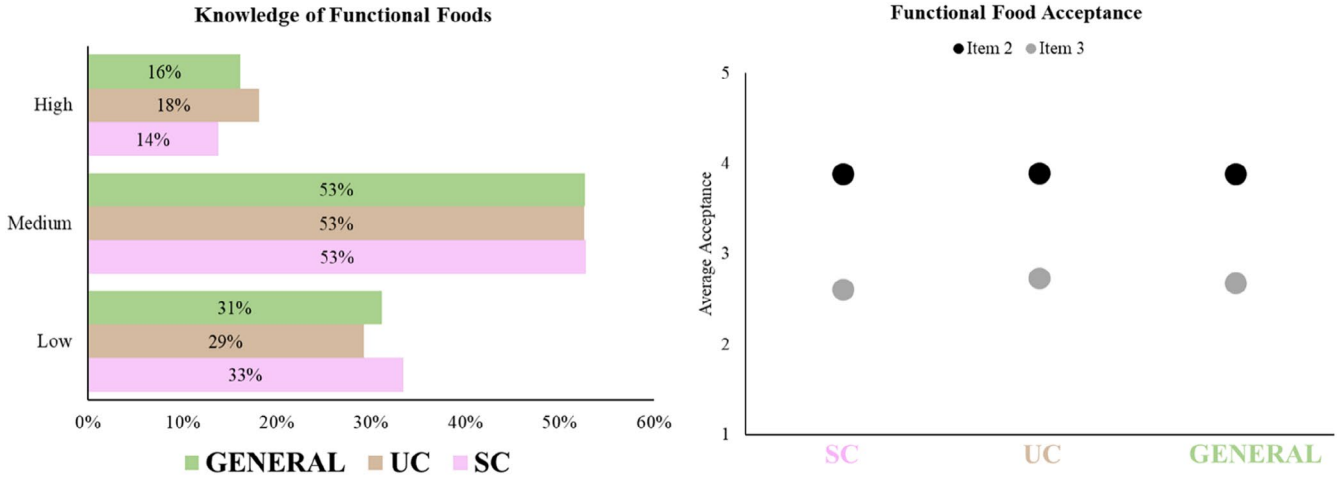


FIGURE 2 | Declaration of knowledge about functional foods and acceptance of functional foods of research participants. Overall and by groups: Consumers of sweetened coffee (SC) and unsweetened coffee (UC) beverages. Items 2 ("Functional foods are all right for me as long as they taste good") and 3 ("Functional foods are all right for me even if they taste worse than their conventional counterpart foods") were evaluated on a 5-point Likert scale (1 = totally disagree, and 5 = totally agree).

Given the importance of reduced sugar consumption, comparing the profiles of the two groups can provide more detailed insight into which characteristics are associated with this behavioral variation.

Focusing on sociodemographic characteristics, the UC group showed a higher prevalence of individuals aged 25–55 (73%) and those with a family income exceeding five times the minimum wages (73%). In contrast, the SC group included a greater proportion of younger participants (15%) and individuals over 55 years old (22%), as well as a higher concentration of participants with lower educational attainment (26% had completed only high school) (Table 4).

In the UC group, a smaller proportion of participants identified themselves as part of the general public (25%), while another share reported working within the coffee chain, such as producers, researchers, traders, or baristas (14%) (Table 4). This professional involvement likely contributes to the higher level of knowledge observed regarding coffee species and varieties, with only 13% of respondents reporting that they did not know the type they consumed (Table 7). UC individuals were also characterized by a higher consumption of pure coffee (89%), a greater frequency of consumption in out-of-home settings such as coffee shops (Table 7), and a higher overall daily intake: 78% reported drinking coffee two or more times per day (Table 6). This group, therefore, demonstrates greater familiarity and interest in the product.

Upon closer examination of behavioral aspects, 66% of all study participants were classified as neutral. However, a comparatively higher proportion of SC participants (21%) were categorized as neophobic, whereas 22% of UC participants were identified as neophilic (Table 9). This contrast suggests a potential shift in consumer behavior, with SC participants possibly transitioning from neophobia to a more neutral stance, while UC participants may be evolving from neutrality toward greater neophilia.

Regarding the acceptance of functional foods, both segments demonstrated similar patterns. On a 5-point Likert scale, the average scores for item 2 were 3.88 (SC) and 3.89 (UC) ($p = 0.782$), and for item 3, 2.60 (SC) and 2.73 (UC) ($p = 0.055$), indicating no significant differences. Nevertheless, UC participants reported greater awareness of functional foods (18%) (Figure 2) and a higher level of knowledge about the health benefits of coffee consumption (74%) (Table 6). In a recent Brazilian survey, in which a high proportion of UC coffee beverage consumers (51%), 73% of the participants believe that coffee consumption has more health benefits than risks (Universidade do Café 2025).

Due to their greater knowledge and interest in the product, as well as lower levels of food neophobia, UC participants appear

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to exhibit more innovative and exploratory consumption behaviors. As described previously, filtered and espresso (machine and capsule) coffee types were the most consumed in both segments. However, UC participants reported a greater diversification in preparation methods, including techniques such as the Italian moka, French press, and Hario V60 (Table 7). While traditional coffee remained the most cited type for home preparation, 57% of UC participants also reported consuming gourmet and superior quality coffees, compared to 29% in the SC group (Table 7).

Although both segments primarily purchase roasted and ground coffee in markets and supermarkets, UC individuals more frequently reported purchasing roasted beans (28%) and capsules (27%), particularly through online platforms (27%) and specialty coffee shops (16%) (Table 7). Moreover, UC participants more actively sought and paid attention to label information regarding roasting degree (47%), quality seals (57%), coffee category (46%), indication of origin (37%), producing region (36%), and sustainability practices (29%) (Table 7).

Altogether, these findings indicate a more engaged and discerning approach among UC participants toward coffee consumption, reflecting a greater openness to innovation and a stronger alignment with current trends in the specialty coffee market. This reinforces the importance of tailoring communication strategies and product offerings to distinct consumer profiles, especially when promoting functional attributes and sustainable practices in the coffee industry.

In summary, therefore, this work demonstrated significant transformations in coffee consumption patterns, reflected through the lens of Brazil, one of the most influential actors in the international coffee trade. Traditional habits, such as drinking coffee at home and associating it with family and routine, continue to shape behavior, while new dynamics reveal a market increasingly influenced by differentiation, health awareness, and sustainability. Notably, the growing pursuit of quality and authenticity among consumers is reflected in the rise of espresso and specialty coffees, the preference for medium roasts, and the increasing importance of certifications. Meanwhile, the growing frequency of going to coffee shops and online sales reflects the modernization of consumption habits, with technology and convenience reshaping access to the product. Despite this evolution, a substantial portion of consumers is still unaware of coffee's functional and health benefits, underscoring the need for improved communication strategies within the industry.

4 | Conclusions

This study outlined emerging trends in coffee consumption in Brazil, which, as one of the world's leading coffee markets, reflects global patterns of increasingly conscious and diversified habits. Deeply cultural yet evolving, coffee remains central to Brazilians' routines, with rising demand for espresso, gourmet, and certified coffees, and a shift toward UC beverages. Although enthusiasm is high, knowledge about health benefits remains limited. UC coffee consumers show greater concern for quality and sustainability, signaling informed and selective consumption. These findings can provide opportunities for the coffee sector and related industries to improve labeling strategies and educational campaigns to promote products with reduced sugar and to increase awareness of the benefits of regular coffee consumption.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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